

# Module 3 DataBase Integration

## Hands on -6: Python ORM

### Crud.py:

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2026-06-21 17:56:41,445 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,446 INFO sqlalchemy.engine.Engine INSERT INTO departments (dept_name, hod_name, budget) VALUES %(dept_name)s, %(hod_name)s, %(budget)s
2026-06-21 17:56:41,447 INFO sqlalchemy.engine.Engine [generated in 0.00019s] {'dept_name': 'Computer Science', 'hod_name': 'Dr Abdul', 'budget': Decimal('50000.00')}
2026-06-21 17:56:41,449 INFO sqlalchemy.engine.Engine INSERT INTO departments (dept_name, hod_name, budget) VALUES %(dept_name)s, %(hod_name)s, %(budget)s
2026-06-21 17:56:41,449 INFO sqlalchemy.engine.Engine [cached since 0.002747s ago] {'dept_name': 'Electrical Engineering', 'hod_name': 'Dr Mani', 'budget': Decimal('400000.00')}
2026-06-21 17:56:41,450 INFO sqlalchemy.engine.Engine INSERT INTO departments (dept_name, hod_name, budget) VALUES %(dept_name)s, %(hod_name)s, %(budget)s
2026-06-21 17:56:41,450 INFO sqlalchemy.engine.Engine [cached since 0.003711s ago] {'dept_name': 'Mechanical Engineering', 'hod_name': 'Dr Ram', 'budget': Decimal('350000.00')}
2026-06-21 17:56:41,452 INFO sqlalchemy.engine.Engine COMMIT
2026-06-21 17:56:41,454 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,455 INFO sqlalchemy.engine.Engine SELECT departments.department_id AS departments_department_id, departments.dept_name AS departments_dept_name, departments.hod_name AS departments_hod_name, departments.budget AS departments_budget
FROM departments
WHERE departments.department_id = %(pk_1)s
2026-06-21 17:56:41,455 INFO sqlalchemy.engine.Engine [generated in 0.00019s] {'pk_1': 1}
2026-06-21 17:56:41,456 INFO sqlalchemy.engine.Engine SELECT departments.department_id AS departments_department_id, departments.dept_name AS departments_dept_name, departments.hod_name AS departments_hod_name, departments.budget AS departments_budget
FROM departments
WHERE departments.department_id = %(pk_1)s
2026-06-21 17:56:41,457 INFO sqlalchemy.engine.Engine [cached since 0.001413s ago] {'pk_1': 2}
2026-06-21 17:56:41,457 INFO sqlalchemy.engine.Engine SELECT departments.department_id AS departments_department_id, departments.dept_name AS departments_dept_name, departments.hod_name AS departments_hod_name, departments.budget AS departments_budget
FROM departments
WHERE departments.department_id = %(pk_1)s
2026-06-21 17:56:41,458 INFO sqlalchemy.engine.Engine [cached since 0.002423s ago] {'pk_1': 3}
2026-06-21 17:56:41,459 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES %(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s
2026-06-21 17:56:41,459 INFO sqlalchemy.engine.Engine [generated in 0.00017s] {'first_name': 'Akash', 'last_name': 'Jagan', 'email': 'akash@gmail.com', 'date_of_birth': datetime.date(2000, 1, 1), 'department_id': 1, 'hiring_year': 2024}
2026-06-21 17:56:41,461 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES %(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s
2026-06-21 17:56:41,461 INFO sqlalchemy.engine.Engine [cached since 0.001754s ago] {'first_name': 'Baru', 'last_name': 'Singh', 'email': 'baru@company.com', 'date_of_birth': datetime.date(1995, 5, 12), 'department_id': 1, 'hiring_year': 2022}
2026-06-21 17:56:41,461 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES %(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s
2026-06-21 17:56:41,461 INFO sqlalchemy.engine.Engine [cached since 0.00239s ago] {'first_name': 'Chandar', 'last_name': 'Babu', 'email': 'chandar@company.com', 'date_of_birth': datetime.date(1998, 8, 20), 'department_id': 2, 'hiring_year': 2024}
2026-06-21 17:56:41,462 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES %(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s
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2026-06-21 17:56:41,477 INFO sqlalchemy.engine.Engine SELECT skill_modules.module_id AS skill_modules_module_id, skill_modules.module_name AS skill_modules_module_name, skill_modules.module_code AS skill_modules_module_code, skill_modules.credits AS skill_modules_credits, skill_modules.department_id AS skill_modules_department_id
FROM skill_modules
WHERE skill_modules.module_id = %(pk_1)s
2026-06-21 17:56:41,477 INFO sqlalchemy.engine.Engine [generated in 0.00015s] {'pk_1': 1}
2026-06-21 17:56:41,478 INFO sqlalchemy.engine.Engine SELECT employees.employee_id AS employees_employee_id, employees.first_name AS employees_first_name, employees.last_name AS employees_last_name, employees.email AS employees_email, employees.date_of_birth AS employees_date_of_birth, employees.department_id AS employees_department_id, employees.hiring_year AS employees_hiring_year
FROM employees
WHERE employees.employee_id = %(pk_1)s
2026-06-21 17:56:41,478 INFO sqlalchemy.engine.Engine [cached since 0.003367s ago] {'pk_1': 2}
2026-06-21 17:56:41,479 INFO sqlalchemy.engine.Engine SELECT skill_modules.module_id AS skill_modules_module_id, skill_modules.module_name AS skill_modules_module_name, skill_modules.module_code AS skill_modules_module_code, skill_modules.credits AS skill_modules_credits, skill_modules.department_id AS skill_modules_department_id
FROM skill_modules
WHERE skill_modules.module_id = %(pk_1)s
2026-06-21 17:56:41,479 INFO sqlalchemy.engine.Engine [cached since 0.002447s ago] {'pk_1': 2}
2026-06-21 17:56:41,480 INFO sqlalchemy.engine.Engine SELECT employees.employee_id AS employees_employee_id, employees.first_name AS employees_first_name, employees.last_name AS employees_last_name, employees.email AS employees_email, employees.date_of_birth AS employees_date_of_birth, employees.department_id AS employees_department_id, employees.hiring_year AS employees_hiring_year
FROM employees
WHERE employees.employee_id = %(pk_1)s
2026-06-21 17:56:41,480 INFO sqlalchemy.engine.Engine [cached since 0.005163s ago] {'pk_1': 3}
2026-06-21 17:56:41,482 INFO sqlalchemy.engine.Engine SELECT skill_modules.module_id AS skill_modules_module_id, skill_modules.module_name AS skill_modules_module_name, skill_modules.module_code AS skill_modules_module_code, skill_modules.credits AS skill_modules_credits, skill_modules.department_id AS skill_modules_department_id
FROM skill_modules
WHERE skill_modules.module_id = %(pk_1)s
2026-06-21 17:56:41,482 INFO sqlalchemy.engine.Engine [cached since 0.004984s ago] {'pk_1': 3}
2026-06-21 17:56:41,484 INFO sqlalchemy.engine.Engine INSERT INTO certifications (employee_id, module_id, completion_date, grade) VALUES %(employee_id)s, %(module_id)s, %(completion_date)s, %(grade)s
2026-06-21 17:56:41,484 INFO sqlalchemy.engine.Engine [generated in 0.00034s] {'employee_id': 1, 'module_id': 1, 'completion_date': datetime.date(2026, 6, 21), 'grade': 'A'}
2026-06-21 17:56:41,486 INFO sqlalchemy.engine.Engine INSERT INTO certifications (employee_id, module_id, completion_date, grade) VALUES %(employee_id)s, %(module_id)s, %(completion_date)s, %(grade)s
2026-06-21 17:56:41,486 INFO sqlalchemy.engine.Engine [cached since 0.002131s ago] {'employee_id': 2, 'module_id': 1, 'completion_date': datetime.date(2026, 6, 21), 'grade': 'B'}
2026-06-21 17:56:41,486 INFO sqlalchemy.engine.Engine INSERT INTO certifications (employee_id, module_id, completion_date, grade) VALUES %(employee_id)s, %(module_id)s, %(completion_date)s, %(grade)s
2026-06-21 17:56:41,487 INFO sqlalchemy.engine.Engine [cached since 0.002756s ago] {'employee_id': 1, 'module_id': 2, 'completion_date': datetime.date(2026, 6, 21), 'grade': 'A'}
2026-06-21 17:56:41,487 INFO sqlalchemy.engine.Engine INSERT INTO certifications (employee_id, module_id, completion_date, grade) VALUES %(employee_id)s, %(module_id)s, %(completion_date)s, %(grade)s
2026-06-21 17:56:41,487 INFO sqlalchemy.engine.Engine [cached since 0.003578s ago] {'employee_id': 3, 'module_id': 3, 'completion_date': datetime.date(2026, 6, 21), 'grade': 'C'}
2026-06-21 17:56:41,488 INFO sqlalchemy.engine.Engine COMMIT
2026-06-21 17:56:41,490 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,491 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
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2026-06-21 17:56:41,462 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES (%(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s)
2026-06-21 17:56:41,462 INFO sqlalchemy.engine.Engine [cached since 0.003144s ago] {'first_name': 'Dee', 'last_name': 'Babu', 'email': 'dee@company.com', 'date_of_birth': datetime.date(1993, 11, 2), 'department_id': 3, 'hiring_year': 2021}
2026-06-21 17:56:41,463 INFO sqlalchemy.engine.Engine INSERT INTO employees (first_name, last_name, email, date_of_birth, department_id, hiring_year) VALUES (%(first_name)s, %(last_name)s, %(email)s, %(date_of_birth)s, %(department_id)s, %(hiring_year)s)
2026-06-21 17:56:41,463 INFO sqlalchemy.engine.Engine [cached since 0.003787s ago] {'first_name': 'Ganesh', 'last_name': 'Kumar', 'email': 'ganesh@company.com', 'date_of_birth': datetime.date(1997, 3, 15), 'department_id': 1, 'hiring_year': 2023}
2026-06-21 17:56:41,463 INFO sqlalchemy.engine.Engine COMMIT
2026-06-21 17:56:41,466 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,466 INFO sqlalchemy.engine.Engine SELECT departments.department_id AS departments_department_id, departments.dept_name AS departments_dept_name, departments.hod_name AS departments_hod_name, departments.budget AS departments_budget
FROM departments
WHERE departments.department_id = %(pk_1)s
2026-06-21 17:56:41,466 INFO sqlalchemy.engine.Engine [cached since 0.01108s ago] {'pk_1': 1}
2026-06-21 17:56:41,467 INFO sqlalchemy.engine.Engine SELECT departments.department_id AS departments_department_id, departments.dept_name AS departments_dept_name, departments.hod_name AS departments_hod_name, departments.budget AS departments_budget
FROM departments
WHERE departments.department_id = %(pk_1)s
2026-06-21 17:56:41,468 INFO sqlalchemy.engine.Engine [cached since 0.01245s ago] {'pk_1': 2}
2026-06-21 17:56:41,471 INFO sqlalchemy.engine.Engine INSERT INTO skill_modules (module_name, module_code, credits, department_id) VALUES (%(module_name)s, %(module_code)s, %(credits)s, %(department_id)s)
2026-06-21 17:56:41,470 INFO sqlalchemy.engine.Engine [generated in 0.00024s] {'module_name': 'Introduction to NoSQL', 'module_code': 'CS101', 'credits': 4, 'department_id': 1}
2026-06-21 17:56:41,471 INFO sqlalchemy.engine.Engine INSERT INTO skill_modules (module_name, module_code, credits, department_id) VALUES (%(module_name)s, %(module_code)s, %(credits)s, %(department_id)s)
2026-06-21 17:56:41,471 INFO sqlalchemy.engine.Engine [cached since 0.001624s ago] {'module_name': 'Microservices Architecture', 'module_code': 'CS102', 'credits': 4, 'department_id': 1}
2026-06-21 17:56:41,471 INFO sqlalchemy.engine.Engine INSERT INTO skill_modules (module_name, module_code, credits, department_id) VALUES (%(module_name)s, %(module_code)s, %(credits)s, %(department_id)s)
2026-06-21 17:56:41,472 INFO sqlalchemy.engine.Engine [cached since 0.002248s ago] {'module_name': 'Circuit Analysis', 'module_code': 'EE201', 'credits': 3, 'department_id': 2}
2026-06-21 17:56:41,472 INFO sqlalchemy.engine.Engine COMMIT
2026-06-21 17:56:41,474 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,475 INFO sqlalchemy.engine.Engine SELECT employees.employee_id AS employees_employee_id, employees.first_name AS employees_first_name, employees.last_name AS employees_last_name, employees.email AS employees_email, employees.date_of_birth AS employees_date_of_birth, employees.department_id AS employees_department_id, employees.hiring_year AS employees_hiring_year
FROM employees
WHERE employees.employee_id = %(pk_1)s
2026-06-21 17:56:41,475 INFO sqlalchemy.engine.Engine [generated in 0.00019s] {'pk_1': 1}
2026-06-21 17:56:41,477 INFO sqlalchemy.engine.Engine SELECT skill_modules.module_id AS skill_modules_module_id, skill_modules.module_name AS skill_modules_module_name, skill_modules.module_code AS ski

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.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
WHERE certifications.certification_id = %(pk_1)s
2026-06-21 17:56:41,491 INFO sqlalchemy.engine.Engine [generated in 0.00019s] {'pk_1': 3}
2026-06-21 17:56:41,493 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
WHERE certifications.certification_id = %(certification_id_1)s
LIMIT %(param_1)s
2026-06-21 17:56:41,493 INFO sqlalchemy.engine.Engine [generated in 0.00027s] {'certification_id_1': 3, 'param_1': 1}
2026-06-21 17:56:41,495 INFO sqlalchemy.engine.Engine DELETE FROM certifications WHERE certifications.certification_id = %(certification_id)s
2026-06-21 17:56:41,495 INFO sqlalchemy.engine.Engine [generated in 0.00022s] {'certification_id': 3}
2026-06-21 17:56:41,495 INFO sqlalchemy.engine.Engine COMMIT
3 deleted
2026-06-21 17:56:41,497 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-21 17:56:41,498 INFO sqlalchemy.engine.Engine SELECT employees.employee_id AS employees_employee_id, employees.first_name AS employees_first_name, employees.last_name AS employees_last_name, employees.email AS employees_email, employees.date_of_birth AS employees_date_of_birth, employees.department_id AS employees_department_id, employees.hiring_year AS employees_hiring_year
FROM employees INNER JOIN departments ON departments.department_id = employees.department_id
WHERE departments.dept_name = %(dept_name_1)s
2026-06-21 17:56:41,499 INFO sqlalchemy.engine.Engine [generated in 0.00031s] {'dept_name_1': 'Computer Science'}
N+1 loading
2026-06-21 17:56:41,501 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
WHERE %(param_1)s = certifications.employee_id
2026-06-21 17:56:41,502 INFO sqlalchemy.engine.Engine [generated in 0.00034s] {'param_1': 1}
Name: Akash Jagan, Course: [models.Certification object at 0x00000294A31F41F0]
2026-06-21 17:56:41,503 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
WHERE %(param_1)s = certifications.employee_id
2026-06-21 17:56:41,503 INFO sqlalchemy.engine.Engine [cached since 0.00182s ago] {'param_1': 2}
Name: Banu Singh, Course: [models.Certification object at 0x00000294A31F4640]
2026-06-21 17:56:41,504 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
WHERE %(param_1)s = certifications.employee_id

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WHERE %(param_1)s = certifications.employee_id
2026-06-21 17:56:41,504 INFO sqlalchemy.engine.Engine [cached since 0.002782s ago] {'param_1': 5}
Name: Ganesh Kumar, Course: []
2026-06-21 17:56:41,505 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade
FROM certifications
2026-06-21 17:56:41,505 INFO sqlalchemy.engine.Engine [generated in 0.00013s] {}
Certifications
<models.Certification object at 0x00000294A31F41F0>
<models.Certification object at 0x00000294A31F4640>
<models.Certification object at 0x00000294A31F46C0>
Eager loading
2026-06-21 17:56:41,509 INFO sqlalchemy.engine.Engine SELECT certifications.certification_id AS certifications_certification_id, certifications.employee_id AS certifications_employee_id, certifications.module_id AS certifications_module_id, certifications.completion_date AS certifications_completion_date, certifications.grade AS certifications_grade, employees_1.employee_id AS employees_1_employee_id, employees_1.first_name AS employees_1_first_name, employees_1.last_name AS employees_1_last_name, employees_1.email AS employees_1_email, employees_1.date_of_birth AS employees_1_date_of_birth, employees_1.department_id AS employees_1_department_id, employees_1.hiring_year AS employees_1_hiring_year, skill_modules_1.module_id AS skill_modules_1_module_id, skill_modules_1.module_name AS skill_modules_1_module_name, skill_modules_1.module_code AS skill_modules_1_module_code, skill_modules_1.credits AS skill_modules_1_credits, skill_modules_1.department_id AS skill_modules_1_department_id
FROM certifications LEFT OUTER JOIN employees AS employees_1 ON employees_1.employee_id = certifications.employee_id LEFT OUTER JOIN skill_modules AS skill_modules_1 ON skill_modules_1.module_id = certifications.module_id
2026-06-21 17:56:41,509 INFO sqlalchemy.engine.Engine [generated in 0.00028s] {}
<models.Certification object at 0x00000294A31F6A00>
<models.Certification object at 0x00000294A31F6A70>
<models.Certification object at 0x00000294A31F6B60>
2026-06-21 17:56:41,511 INFO sqlalchemy.engine.Engine SELECT employees.employee_id AS employees_employee_id, employees.first_name AS employees_first_name, employees.last_name AS employees_last_name, employees.email AS employees_email, employees.date_of_birth AS employees_date_of_birth, employees.department_id AS employees_department_id, employees.hiring_year AS employees_hiring_year
FROM employees
WHERE employees.email = %(email_1)s
LIMIT %(param_1)s
2026-06-21 17:56:41,512 INFO sqlalchemy.engine.Engine [generated in 0.00020s] {'email_1': 'dee@company.com', 'param_1': 1}
2026-06-21 17:56:41,513 INFO sqlalchemy.engine.Engine UPDATE employees SET hiring_year=%(hiring_year)s WHERE employees.employee_id = %(employees_employee_id)s
2026-06-21 17:56:41,513 INFO sqlalchemy.engine.Engine [generated in 0.00021s] {'hiring_year': 2026, 'employees_employee_id': 4}
2026-06-21 17:56:41,514 INFO sqlalchemy.engine.Engine COMMIT

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